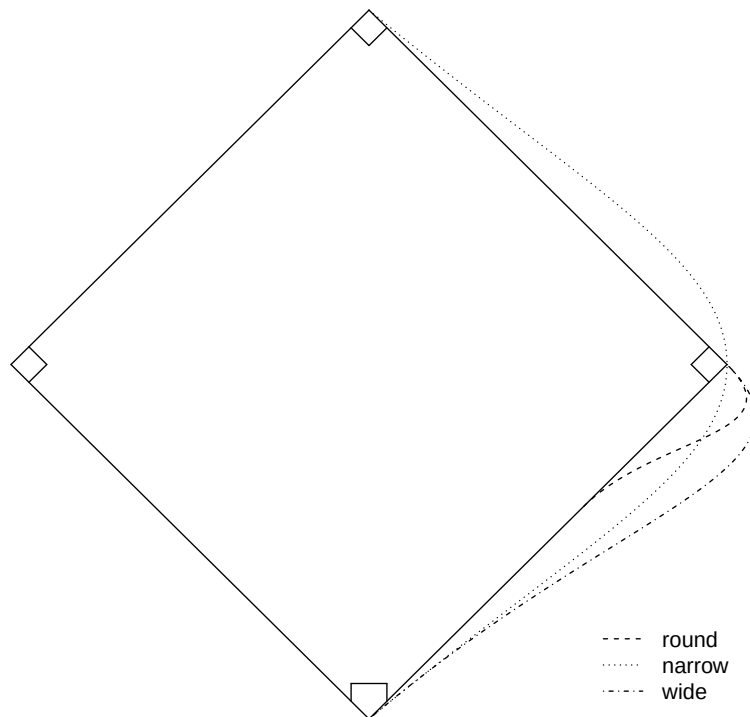


Friday, April 18

The Incidental Parameter Problem

Some kinds of designs result in a “factor” with a relatively large number of levels, where each level corresponds to an experimental/observational unit. This can arise for a variety of reasons. Such designs include *repeated measures*, *longitudinal data*, *panel data*, *multilevel data*, *pseudo-replication*, *within-subjects factors*, *dependent samples*, and *clustered data* to name a few (these are not mutually exclusive). Having a factor with a large number of levels can cause complications. This is known in econometrics as the “incidental parameter problem.”

Example: Consider a study of the running times of three routes from home to second base on a baseball diamond.



```
library(trtools)
head(baserun)
```

	round	narrow	wide
1	5.40	5.50	5.55
2	5.85	5.70	5.75
3	5.20	5.60	5.50
4	5.55	5.50	5.40
5	5.90	5.85	5.70
6	5.45	5.55	5.60

There is a considerable “effect” for the player. Players who are relatively fast/slow on one route tend to also be relatively fast/slow on the other routes.

```

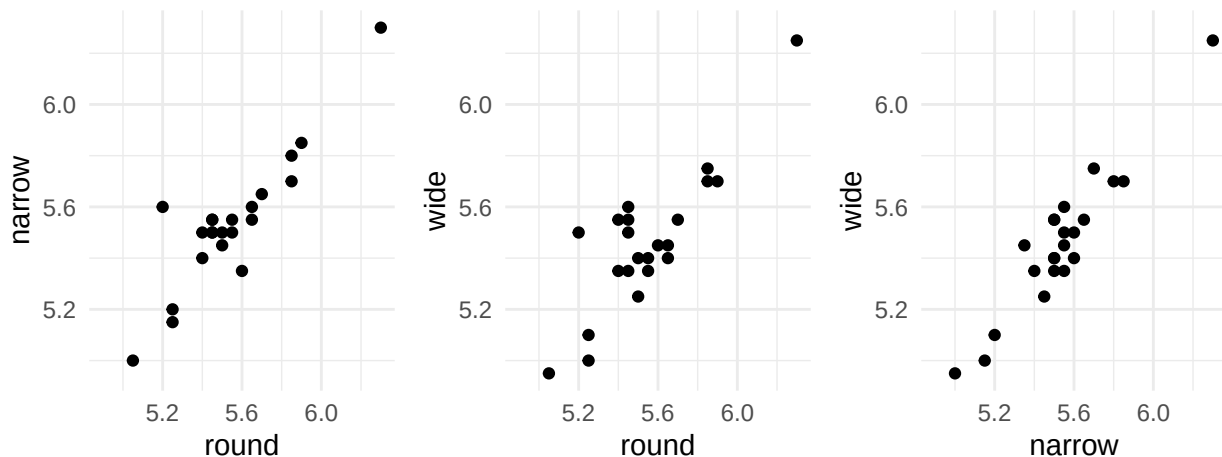
p <- ggplot(baserun, aes(x = round, y = narrow)) + theme_minimal() +
  geom_point() + xlim(4.9,6.3) + ylim(4.95,6.3)
p1 <- p

p <- ggplot(baserun, aes(x = round, y = wide)) + theme_minimal() +
  geom_point() + xlim(4.9,6.3) + ylim(4.95,6.3)
p2 <- p

p <- ggplot(baserun, aes(x = narrow, y = wide)) + theme_minimal() +
  geom_point() + xlim(4.9,6.3) + ylim(4.95,6.3)
p3 <- p

cowplot::plot_grid(p1, p2, p3, align = "h", ncol = 3)

```



These data are in what is sometimes called “wide form” where there are multiple observations per unit (player) in a single row. For plotting and modeling it is often useful to “reshape” the data into “long form” with one observation of the response variable (running time) per row.

```

library(dplyr)
library(tidyr)
baselong <- baserun |> mutate(player = factor(letters[1:n()]))) |>
  pivot_longer(cols = c(round, narrow, wide),
    names_to = "route", values_to = "time")
head(baselong)

```

```

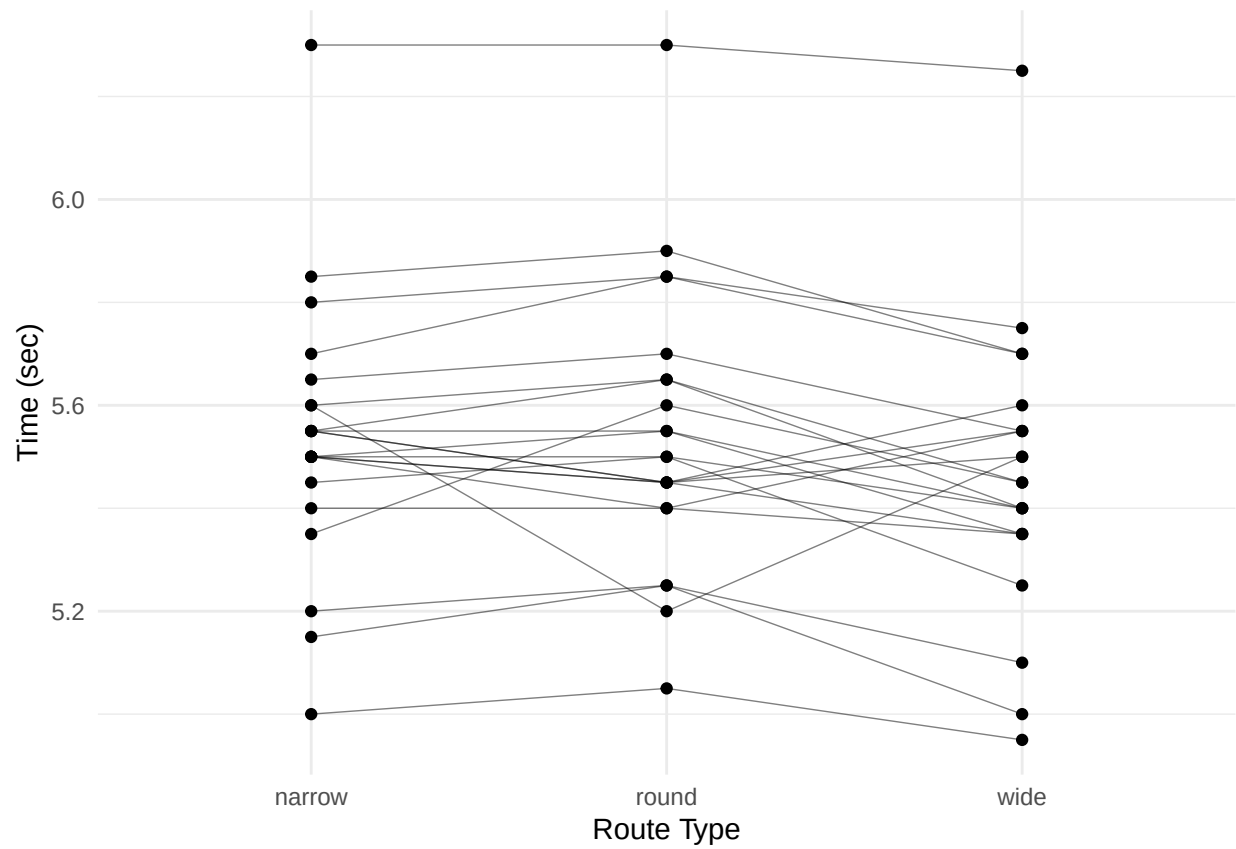
# A tibble: 6 x 3
  player route  time
  <fct> <chr> <dbl>
1 a     round  5.4
2 a     narrow  5.5
3 a     wide   5.55
4 b     round  5.85
5 b     narrow  5.7
6 b     wide   5.75

```

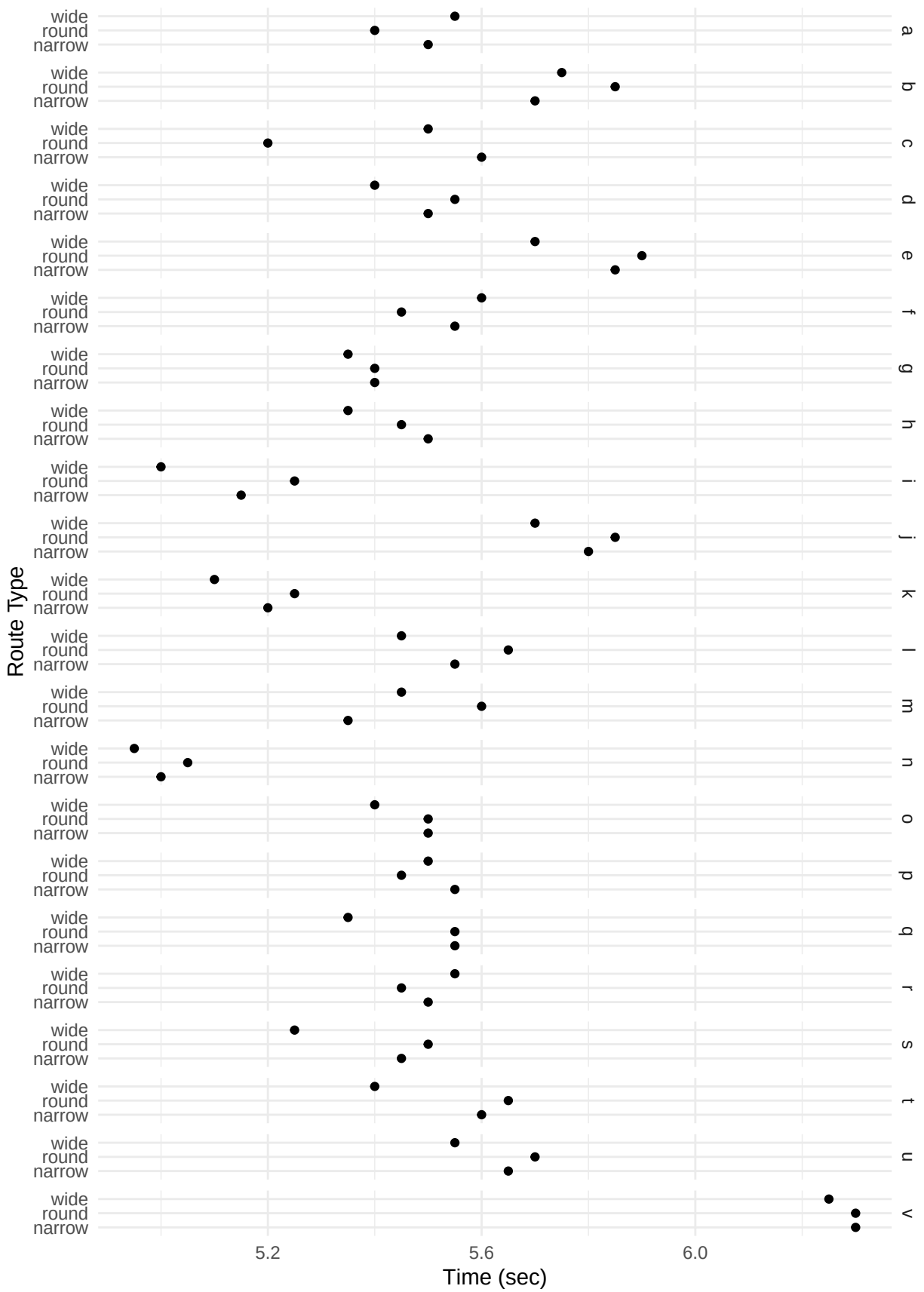
```

p <- ggplot(baselong, aes(x = route, y = time)) +
  geom_line(aes(group = player), linewidth = 0.25, alpha = 0.5) +
  geom_point() + theme_minimal() +
  labs(x = "Route Type", y = "Time (sec)")
plot(p)

```



```
p <- ggplot(baselong, aes(x = time, y = route)) +
  geom_point() + theme_minimal() + facet_grid(player ~ .) +
  labs(y = "Route Type", x = "Time (sec)")
plot(p)
```



Again note that there appears to be a “player effect” in that the players show similar results over the routes.

What *could* we do (but not necessarily what we *should* do) in modeling these data.

We could ignore the effect of player.

```
m <- lm(time ~ route, data = baselong)
summary(m)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.53409	0.0572	96.784	3.05e-70
routeround	0.00909	0.0809	0.112	9.11e-01
routewide	-0.07500	0.0809	-0.927	3.57e-01

Or we could model the effect of player as a factor.

```
m <- lm(time ~ route + player, data = baselong)
summary(m)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.51e+00	0.0521	1.06e+02	1.32e-52
routeround	9.09e-03	0.0260	3.49e-01	7.29e-01
routewide	-7.50e-02	0.0260	-2.88e+00	6.21e-03
playerb	2.83e-01	0.0705	4.02e+00	2.37e-04
playerc	-5.00e-02	0.0705	-7.09e-01	4.82e-01
playerd	1.18e-15	0.0705	1.67e-14	1.00e+00
playere	3.33e-01	0.0705	4.73e+00	2.55e-05
playerf	5.00e-02	0.0705	7.09e-01	4.82e-01
playerg	-1.00e-01	0.0705	-1.42e+00	1.63e-01
playerh	-5.00e-02	0.0705	-7.09e-01	4.82e-01
playeri	-3.50e-01	0.0705	-4.97e+00	1.19e-05
playerj	3.00e-01	0.0705	4.26e+00	1.14e-04
playerk	-3.00e-01	0.0705	-4.26e+00	1.14e-04
playerl	6.67e-02	0.0705	9.46e-01	3.50e-01
playerm	-1.67e-02	0.0705	-2.36e-01	8.14e-01
playern	-4.83e-01	0.0705	-6.86e+00	2.32e-08
playero	-1.67e-02	0.0705	-2.36e-01	8.14e-01
playerp	1.67e-02	0.0705	2.36e-01	8.14e-01
playerq	8.79e-16	0.0705	1.25e-14	1.00e+00
playerr	1.67e-02	0.0705	2.36e-01	8.14e-01
players	-8.33e-02	0.0705	-1.18e+00	2.44e-01
playert	6.67e-02	0.0705	9.46e-01	3.50e-01
playeru	1.50e-01	0.0705	2.13e+00	3.92e-02
playerv	8.00e-01	0.0705	1.14e+01	2.24e-14

Or maybe we could do something else?

Example: Consider the following data from a meta-analysis of 26 studies of the effect of nicotine gum on smoking cessation.

```
library(HSAUR3) # for the data
head(smoking)
```

	qt	tt	qc	tc
Blondal89	37	92	24	90
Campbell191	21	107	21	105
Fagerstrom82	30	50	23	50
Fee82	23	180	15	172
Garcia89	21	68	5	38

Garvey00 75 405 17 203

Here `qt` and `tc` are the total number of subjects in the treatment and control groups, respectively, and `tt` and `tc` are the total number of subjects in the treatment and control groups, respectively.

These data require some rearranging prior to plotting and analysis. (Note: I'm using `dplyr::select` rather than just `select` because of a conflict with a function of the same name with another package I have loaded.)

```
library(dplyr)
library(tidyr)
quitsmoke <- smoking
quitsmoke$study <- rownames(quitsmoke)
quitsmoke.quits <- quitsmoke |> dplyr::select(study, qt, qc) |>
  rename(gum = qt, control = qc) |>
  pivot_longer(cols = c(gum, control),
    names_to = "treatment", values_to = "quit")
head(quitsmoke.quits)
```

```
# A tibble: 6 x 3
  study      treatment quit
  <chr>      <chr>    <int>
1 Blondal89  gum         37
2 Blondal89  control     24
3 Campbell191 gum         21
4 Campbell191 control     21
5 Fagerstrom82 gum         30
6 Fagerstrom82 control     23
```

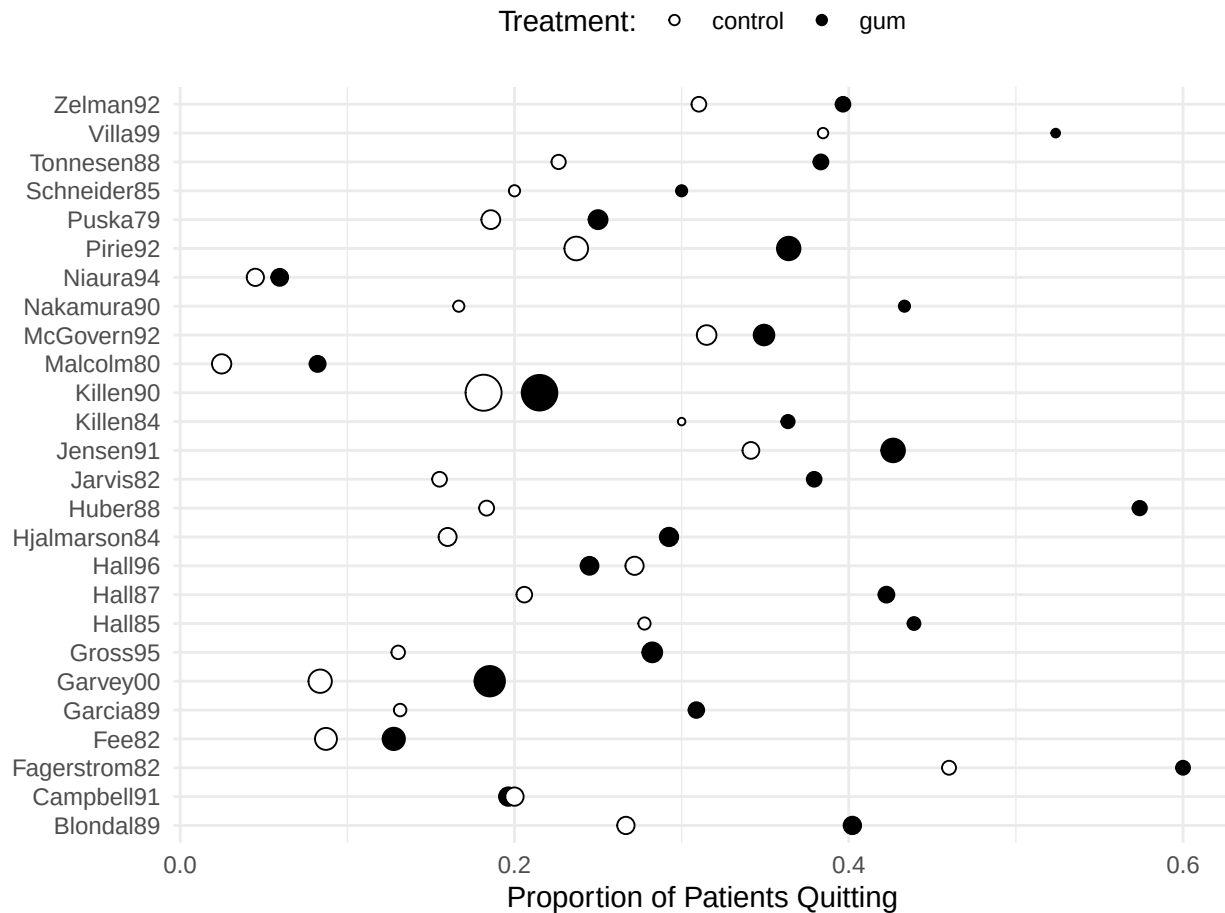
```
quitsmoke.total <- quitsmoke |> dplyr::select(study, tt, tc) |>
  rename(gum = tt, control = tc) |>
  pivot_longer(cols = c(gum, control), names_to = "treatment", values_to = "total")
head(quitsmoke.total)
```

```
# A tibble: 6 x 3
  study      treatment total
  <chr>      <chr>    <int>
1 Blondal89  gum         92
2 Blondal89  control     90
3 Campbell191 gum        107
4 Campbell191 control    105
5 Fagerstrom82 gum         50
6 Fagerstrom82 control     50
```

```
quitsmoke <- full_join(quitsmoke.quits, quitsmoke.total) |> mutate(study = factor(study)) |> arrange(study)
head(quitsmoke)
```

```
# A tibble: 6 x 4
  study      treatment quit total
  <fct>      <chr>    <int> <int>
1 Blondal89  gum         37    92
2 Blondal89  control     24    90
3 Campbell191 gum         21   107
4 Campbell191 control     21   105
5 Fagerstrom82 gum         30    50
6 Fagerstrom82 control     23    50
```

```
p <- ggplot(quitsmoke, aes(x = study, y = quit/total,
  size = total, fill = treatment)) + geom_point(pch = 21) +
  coord_flip() + guides(size = "none") +
  scale_fill_manual(values = c("White", "Black")) + theme_minimal() +
  labs(x = NULL, y = "Proportion of Patients Quitting",
    fill = "Treatment:") + theme(legend.position = "top")
plot(p)
```



The studies may vary considerably in terms of (a) the proportion of subjects that quit overall and (b) the effectiveness of the gum treatment relative to the control condition.

What *could* we do (but not necessarily what we *should* do) in modeling these data.

We could ignore the effect of study.

```
m <- glm(cbind(quit, total - quit) ~ treatment,
  family = binomial, data = quitsmoke)
summary(m)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.450	0.0490	-29.59	1.76e-192
treatmentgum	0.507	0.0631	8.04	9.11e-16

Or we could model the main effect of study.

```
m <- glm(cbind(quit, total - quit) ~ treatment + study,
  family = binomial, data = quitsmoke)
```

```
summary(m)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.9561	0.1622	-5.893	3.78e-09
treatmentgum	0.5148	0.0657	7.834	4.74e-15
studyCampbell91	-0.7218	0.2346	-3.077	2.09e-03
studyFagerstrom82	0.8209	0.2566	3.199	1.38e-03
studyFee82	-1.4447	0.2339	-6.176	6.57e-10
studyGarcia89	-0.5137	0.2768	-1.856	6.35e-02
studyGarvey00	-1.1312	0.1951	-5.797	6.75e-09
studyGross95	-0.5748	0.2372	-2.424	1.54e-02
studyHall85	0.1132	0.2863	0.395	6.93e-01
studyHall87	-0.0887	0.2424	-0.366	7.14e-01
studyHall96	-0.3636	0.2265	-1.605	1.08e-01
studyHjalmarson84	-0.5455	0.2300	-2.372	1.77e-02
studyHuber88	0.1647	0.2516	0.654	5.13e-01
studyJarvis82	-0.3254	0.2638	-1.233	2.17e-01
studyJensen91	0.1852	0.1989	0.931	3.52e-01
studyKillen84	-0.0539	0.3086	-0.175	8.61e-01
studyKillen90	-0.7163	0.1739	-4.119	3.81e-05
studyMalcolm80	-2.2897	0.3767	-6.078	1.21e-09
studyMcGovern92	-0.0235	0.2043	-0.115	9.08e-01
studyNakamura90	-0.1619	0.3248	-0.498	6.18e-01
studyNiaura94	-2.2260	0.3776	-5.894	3.76e-09
studyPirie92	-0.1599	0.1913	-0.836	4.03e-01
studyPuska79	-0.5987	0.2256	-2.654	7.96e-03
studySchneider85	-0.4165	0.3391	-1.228	2.19e-01
studyTonnesen88	-0.1313	0.2588	-0.507	6.12e-01
studyVilla99	0.5093	0.3355	1.518	1.29e-01
studyZelman92	0.0851	0.2516	0.338	7.35e-01

We could also model an interaction of the treatment with the study.

```
m <- glm(cbind(quit, total - quit) ~ treatment * study,
  family = binomial, data = quitsmoke)
summary(m)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.01160	0.238	-4.2439	2.20e-05
treatmentgum	0.61519	0.319	1.9260	5.41e-02
studyCampbell91	-0.37469	0.341	-1.0985	2.72e-01
studyFagerstrom82	0.85126	0.371	2.2971	2.16e-02
studyFee82	-1.33659	0.360	-3.7091	2.08e-04
studyGarcia89	-0.87547	0.536	-1.6338	1.02e-01
studyGarvey00	-1.38093	0.348	-3.9696	7.20e-05
studyGross95	-0.88552	0.498	-1.7764	7.57e-02
studyHall85	0.05609	0.442	0.1269	8.99e-01
studyHall87	-0.33833	0.383	-0.8831	3.77e-01
studyHall96	0.02632	0.325	0.0809	9.36e-01
studyHjalmarson84	-0.64663	0.362	-1.7850	7.43e-02
studyHuber88	-0.48232	0.410	-1.1763	2.39e-01
studyJarvis82	-0.68299	0.434	-1.5738	1.16e-01
studyJensen91	0.35482	0.333	1.0648	2.87e-01
studyKillen84	0.16430	0.543	0.3026	7.62e-01
studyKillen90	-0.49446	0.260	-1.9000	5.74e-02

studyMalcolm80	-2.66047	0.631	-4.2138	2.51e-05
studyMcGovern92	0.23457	0.305	0.7679	4.43e-01
studyNakamura90	-0.59784	0.545	-1.0973	2.72e-01
studyNiaura94	-2.04476	0.564	-3.6227	2.92e-04
studyPirie92	-0.15778	0.288	-0.5476	5.84e-01
studyPuska79	-0.46567	0.340	-1.3713	1.70e-01
studySchneider85	-0.37469	0.515	-0.7277	4.67e-01
studyTonnesen88	-0.21706	0.406	-0.5351	5.93e-01
studyVilla99	0.54160	0.468	1.1565	2.47e-01
studyZelman92	0.21309	0.371	0.5749	5.65e-01
treatmentgum:studyCampbell91	-0.63872	0.470	-1.3593	1.74e-01
treatmentgum:studyFagerstrom82	-0.04938	0.516	-0.0958	9.24e-01
treatmentgum:studyFee82	-0.18774	0.474	-0.3959	6.92e-01
treatmentgum:studyGarcia89	0.46626	0.633	0.7361	4.62e-01
treatmentgum:studyGarvey00	0.29574	0.427	0.6921	4.89e-01
treatmentgum:studyGross95	0.34956	0.576	0.6073	5.44e-01
treatmentgum:studyHall85	0.09520	0.583	0.1634	8.70e-01
treatmentgum:studyHall87	0.42237	0.500	0.8452	3.98e-01
treatmentgum:studyHall96	-0.75591	0.454	-1.6644	9.60e-02
treatmentgum:studyHjalmarson84	0.15954	0.471	0.3386	7.35e-01
treatmentgum:studyHuber88	1.17723	0.538	2.1895	2.86e-02
treatmentgum:studyJarvis82	0.58693	0.554	1.0597	2.89e-01
treatmentgum:studyJensen91	-0.25439	0.419	-0.6070	5.44e-01
treatmentgum:studyKillen84	-0.32750	0.662	-0.4947	6.21e-01
treatmentgum:studyKillen90	-0.40417	0.350	-1.1533	2.49e-01
treatmentgum:studyMalcolm80	0.64395	0.791	0.8143	4.15e-01
treatmentgum:studyMcGovern92	-0.46021	0.411	-1.1206	2.62e-01
treatmentgum:studyNakamura90	0.72599	0.691	1.0503	2.94e-01
treatmentgum:studyNiaura94	-0.31884	0.759	-0.4199	6.75e-01
treatmentgum:studyPirie92	-0.00351	0.386	-0.0091	9.93e-01
treatmentgum:studyPuska79	-0.23653	0.454	-0.5205	6.03e-01
treatmentgum:studySchneider85	-0.07619	0.685	-0.1112	9.11e-01
treatmentgum:studyTonnesen88	0.13806	0.529	0.2608	7.94e-01
treatmentgum:studyVilla99	-0.04987	0.675	-0.0739	9.41e-01
treatmentgum:studyZelman92	-0.23653	0.505	-0.4687	6.39e-01

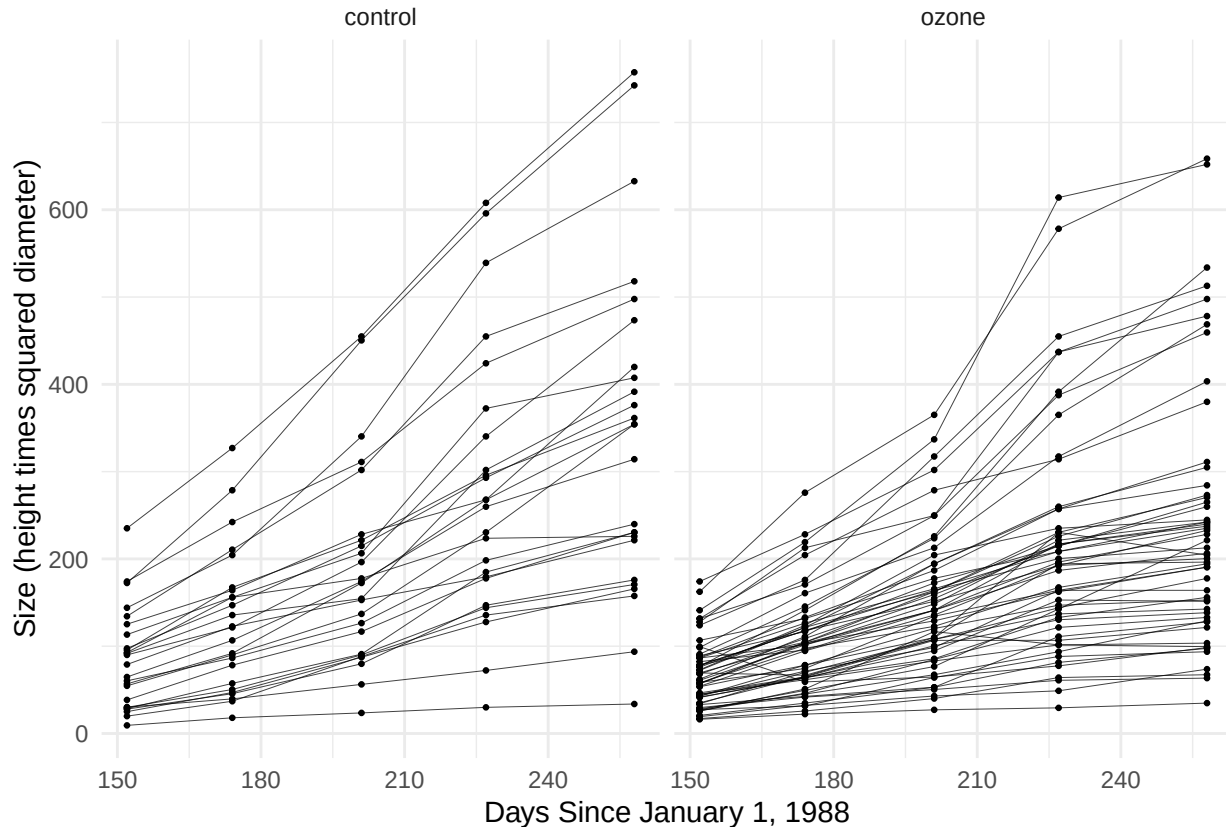
Or maybe we could do something else?

Example: Consider the following data from a study of the growth of Sitka spruce trees under two experimental conditions.

```
library(MASS)
head(Sitka, 10) # note that size is on log scale
```

	size	Time	tree	treat	treesize
1	4.51	152	1	ozone	90.9
2	4.98	174	1	ozone	145.5
3	5.41	201	1	ozone	223.6
4	5.90	227	1	ozone	365.0
5	6.15	258	1	ozone	468.7
6	4.24	152	2	ozone	69.4
7	4.20	174	2	ozone	66.7
8	4.68	201	2	ozone	107.8
9	4.92	227	2	ozone	137.0
10	4.96	258	2	ozone	142.6

```
Sitka$treeshize <- exp(Sitka$size)
p <- ggplot(Sitka, aes(x = Time, y = treeshize)) +
  geom_line(aes(group = tree), alpha = 0.75, linewidth = 0.1) +
  facet_wrap(~ treat) + geom_point(size = 0.5) +
  labs(y = "Size (height times squared diameter)",
       x = "Days Since January 1, 1988") + theme_minimal()
plot(p)
```



Note that trees vary considerably in terms of their growth trajectories.

What *could* we do (but not necessarily what we *should* do) in modeling these data.

We could ignore the effect of tree.

```
m <- lm(treeshize ~ Time * treat, data = Sitka)
summary(m)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-305.123	52.711	-5.79	1.46e-08
Time	2.509	0.256	9.80	2.03e-20
treatozone	110.675	63.755	1.74	8.34e-02
Time:treatozone	-0.788	0.310	-2.54	1.13e-02

Or we could model the effect of tree.

```
Sitka$tree <- factor(Sitka$tree)
m <- lm(treeshize ~ Time * treat + Time * tree, data = Sitka)
summary(m)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
--	----------	------------	---------	----------

(Intercept)	-1.97e+02	48.007	-4.1123	5.41e-05
Time	1.41e+00	0.233	6.0396	5.93e-09
treatozone	-2.91e+02	67.892	-4.2886	2.62e-05
tree2	4.28e+02	67.892	6.3030	1.41e-09
tree3	3.97e+02	67.892	5.8478	1.64e-08
tree4	3.78e+02	67.892	5.5673	6.99e-08
tree5	-1.32e+02	67.892	-1.9382	5.38e-02
tree6	1.41e+02	67.892	2.0738	3.92e-02
tree7	3.72e+02	67.892	5.4802	1.08e-07
tree8	2.97e+02	67.892	4.3739	1.83e-05
tree9	6.93e-01	67.892	0.0102	9.92e-01
tree10	4.33e+02	67.892	6.3744	9.50e-10
tree11	3.81e+02	67.892	5.6068	5.72e-08
tree12	2.50e+02	67.892	3.6849	2.83e-04
tree13	2.47e+02	67.892	3.6451	3.28e-04
tree14	3.65e+02	67.892	5.3794	1.79e-07
tree15	5.51e+02	67.892	8.1198	2.56e-14
tree16	3.86e+02	67.892	5.6921	3.69e-08
tree17	3.97e+02	67.892	5.8423	1.69e-08
tree18	4.36e+02	67.892	6.4158	7.54e-10
tree19	4.14e+02	67.892	6.1024	4.23e-09
tree20	3.51e+02	67.892	5.1690	5.00e-07
tree21	3.70e+02	67.892	5.4475	1.28e-07
tree22	3.21e+02	67.892	4.7231	3.98e-06
tree23	2.70e+02	67.892	3.9806	9.14e-05
tree24	4.81e+02	67.892	7.0840	1.59e-11
tree25	2.20e+02	67.892	3.2440	1.35e-03
tree26	3.69e+02	67.892	5.4406	1.32e-07
tree27	2.63e+02	67.892	3.8725	1.39e-04
tree28	3.24e+02	67.892	4.7655	3.29e-06
tree29	4.93e+01	67.892	0.7256	4.69e-01
tree30	2.90e+02	67.892	4.2711	2.82e-05
tree31	3.63e+02	67.892	5.3397	2.18e-07
tree32	3.19e+02	67.892	4.7010	4.39e-06
tree33	3.23e+02	67.892	4.7548	3.45e-06
tree34	3.56e+02	67.892	5.2467	3.43e-07
tree35	1.63e+02	67.892	2.4006	1.71e-02
tree36	4.55e+02	67.892	6.7025	1.48e-10
tree37	-8.90e+01	67.892	-1.3113	1.91e-01
tree38	1.93e+02	67.892	2.8407	4.89e-03
tree39	1.37e+02	67.892	2.0145	4.51e-02
tree40	3.08e+02	67.892	4.5327	9.24e-06
tree41	-1.97e+02	67.892	-2.9058	4.01e-03
tree42	3.19e+02	67.892	4.7004	4.41e-06
tree43	2.34e+02	67.892	3.4434	6.79e-04
tree44	3.06e+02	67.892	4.5113	1.01e-05
tree45	4.26e+02	67.892	6.2750	1.65e-09
tree46	2.80e+02	67.892	4.1250	5.13e-05
tree47	3.29e+02	67.892	4.8446	2.29e-06
tree48	3.64e+02	67.892	5.3660	1.91e-07
tree49	4.05e+02	67.892	5.9724	8.50e-09
tree50	3.93e+02	67.892	5.7923	2.20e-08
tree51	3.52e+02	67.892	5.1810	4.72e-07
tree52	2.66e+02	67.892	3.9244	1.14e-04

tree53	4.72e+02	67.892	6.9581	3.35e-11
tree54	3.55e+02	67.892	5.2339	3.65e-07
tree55	1.23e+02	67.892	1.8046	7.24e-02
tree56	-4.33e+02	67.892	-6.3798	9.22e-10
tree57	8.88e+01	67.892	1.3076	1.92e-01
tree58	-1.15e+02	67.892	-1.6947	9.15e-02
tree59	-2.00e+02	67.892	-2.9462	3.54e-03
tree60	-1.66e+02	67.892	-2.4433	1.53e-02
tree61	-4.67e+02	67.892	-6.8724	5.53e-11
tree62	-2.06e+01	67.892	-0.3027	7.62e-01
tree63	1.12e+01	67.892	0.1644	8.70e-01
tree64	1.74e+02	67.892	2.5672	1.09e-02
tree65	-4.37e+01	67.892	-0.6433	5.21e-01
tree66	8.09e+00	67.892	0.1192	9.05e-01
tree67	-1.05e+02	67.892	-1.5480	1.23e-01
tree68	-2.05e+02	67.892	-3.0179	2.82e-03
tree69	-1.76e+02	67.892	-2.5978	9.97e-03
tree70	-7.68e+01	67.892	-1.1315	2.59e-01
tree71	-2.49e+02	67.892	-3.6695	3.00e-04
tree72	-9.72e+01	67.892	-1.4317	1.54e-01
tree73	-3.40e+02	67.892	-5.0103	1.06e-06
tree74	-1.16e+02	67.892	-1.7143	8.78e-02
tree75	-9.12e+01	67.892	-1.3429	1.81e-01
tree76	-1.13e+01	67.892	-0.1665	8.68e-01
tree77	1.34e+02	67.892	1.9673	5.03e-02
tree78	-3.18e+02	67.892	-4.6784	4.86e-06
Time:treatozone	2.28e+00	0.330	6.9248	4.07e-11
Time:tree2	-2.88e+00	0.330	-8.7184	5.01e-16
Time:tree3	-2.69e+00	0.330	-8.1702	1.85e-14
Time:tree4	-2.38e+00	0.330	-7.2218	6.95e-12
Time:tree5	7.25e-01	0.330	2.1968	2.90e-02
Time:tree6	-7.95e-01	0.330	-2.4118	1.66e-02
Time:tree7	-2.41e+00	0.330	-7.3159	3.93e-12
Time:tree8	-1.98e+00	0.330	-6.0148	6.77e-09
Time:tree9	2.84e-01	0.330	0.8620	3.90e-01
Time:tree10	-2.98e+00	0.330	-9.0227	6.44e-17
Time:tree11	-2.57e+00	0.330	-7.7950	2.03e-13
Time:tree12	-1.60e+00	0.330	-4.8386	2.36e-06
Time:tree13	-1.54e+00	0.330	-4.6611	5.25e-06
Time:tree14	-2.27e+00	0.330	-6.8839	5.17e-11
Time:tree15	-3.61e+00	0.330	-10.9381	8.24e-23
Time:tree16	-2.72e+00	0.330	-8.2444	1.14e-14
Time:tree17	-2.38e+00	0.330	-7.2213	6.97e-12
Time:tree18	-3.22e+00	0.330	-9.7514	4.17e-19
Time:tree19	-2.93e+00	0.330	-8.8897	1.59e-16
Time:tree20	-2.25e+00	0.330	-6.8180	7.60e-11
Time:tree21	-2.47e+00	0.330	-7.4922	1.34e-12
Time:tree22	-2.34e+00	0.330	-7.0809	1.62e-11
Time:tree23	-1.81e+00	0.330	-5.4800	1.09e-07
Time:tree24	-3.53e+00	0.330	-10.6923	4.96e-22
Time:tree25	-1.86e+00	0.330	-5.6260	5.18e-08
Time:tree26	-2.74e+00	0.330	-8.3207	6.94e-15
Time:tree27	-1.92e+00	0.330	-5.8195	1.91e-08
Time:tree28	-2.03e+00	0.330	-6.1674	2.97e-09

Time:tree29	-7.20e-02	0.330	-0.2184	8.27e-01
Time:tree30	-1.42e+00	0.330	-4.2989	2.51e-05
Time:tree31	-2.59e+00	0.330	-7.8588	1.35e-13
Time:tree32	-2.06e+00	0.330	-6.2604	1.79e-09
Time:tree33	-2.00e+00	0.330	-6.0723	4.97e-09
Time:tree34	-2.41e+00	0.330	-7.2951	4.46e-12
Time:tree35	-4.64e-01	0.330	-1.4075	1.61e-01
Time:tree36	-3.14e+00	0.330	-9.5222	2.07e-18
Time:tree37	1.18e+00	0.330	3.5697	4.33e-04
Time:tree38	-1.31e+00	0.330	-3.9719	9.46e-05
Time:tree39	-5.34e-01	0.330	-1.6194	1.07e-01
Time:tree40	-2.08e+00	0.330	-6.3047	1.40e-09
Time:tree41	1.64e+00	0.330	4.9613	1.34e-06
Time:tree42	-2.07e+00	0.330	-6.2846	1.56e-09
Time:tree43	-1.62e+00	0.330	-4.9060	1.73e-06
Time:tree44	-2.23e+00	0.330	-6.7641	1.04e-10
Time:tree45	-3.17e+00	0.330	-9.6160	1.08e-18
Time:tree46	-1.81e+00	0.330	-5.4984	9.89e-08
Time:tree47	-2.23e+00	0.330	-6.7747	9.77e-11
Time:tree48	-2.72e+00	0.330	-8.2330	1.23e-14
Time:tree49	-2.98e+00	0.330	-9.0370	5.84e-17
Time:tree50	-2.94e+00	0.330	-8.9061	1.42e-16
Time:tree51	-2.61e+00	0.330	-7.9124	9.63e-14
Time:tree52	-2.00e+00	0.330	-6.0662	5.14e-09
Time:tree53	-3.37e+00	0.330	-10.2156	1.55e-20
Time:tree54	-2.54e+00	0.330	-7.6885	3.95e-13
Time:tree55	-1.73e-01	0.330	-0.5241	6.01e-01
Time:tree56	3.55e+00	0.330	10.7499	3.26e-22
Time:tree57	-1.25e-01	0.330	-0.3794	7.05e-01
Time:tree58	1.25e+00	0.330	3.7912	1.90e-04
Time:tree59	1.41e+00	0.330	4.2787	2.73e-05
Time:tree60	1.63e+00	0.330	4.9338	1.52e-06
Time:tree61	4.09e+00	0.330	12.3869	1.67e-27
Time:tree62	3.71e-01	0.330	1.1263	2.61e-01
Time:tree63	-4.08e-02	0.330	-0.1239	9.02e-01
Time:tree64	-1.18e+00	0.330	-3.5808	4.15e-04
Time:tree65	4.17e-01	0.330	1.2633	2.08e-01
Time:tree66	6.26e-03	0.330	0.0190	9.85e-01
Time:tree67	1.72e+00	0.330	5.2026	4.25e-07
Time:tree68	1.51e+00	0.330	4.5816	7.46e-06
Time:tree69	1.50e+00	0.330	4.5506	8.55e-06
Time:tree70	1.07e+00	0.330	3.2580	1.29e-03
Time:tree71	2.40e+00	0.330	7.2715	5.15e-12
Time:tree72	1.24e+00	0.330	3.7532	2.20e-04
Time:tree73	3.60e+00	0.330	10.9203	9.39e-23
Time:tree74	1.05e+00	0.330	3.1855	1.64e-03
Time:tree75	6.03e-01	0.330	1.8269	6.90e-02
Time:tree76	9.33e-02	0.330	0.2829	7.78e-01
Time:tree77	-8.04e-01	0.330	-2.4389	1.55e-02
Time:tree78	2.34e+00	0.330	7.1052	1.40e-11

Or maybe we could do something else?

Marginal Models and Generalized Estimating Equations

A marginal model *ignores* the many-leveled factor. One approach to estimating such models is to use what can be viewed as an extension of quasi-likelihood called *generalized estimating equations* (GEE). This approach actually involves two parts.

1. Estimate the model using generalized estimating equations. This uses an iterative generalized least squares that uses an estimated “working” correlation structure. This can be viewed as an extension of the iteratively weighted least squares algorithm we used earlier.
2. Compute *robust* estimates of standard errors to account for heteroscedasticity and correlations among observations. These are designed to deal with the fact that our observations are not independent.

Example: Consider two approaches to the `baserun` data: ignoring the player effect entirely and a marginal model with inferences based on GEE.

```
library(geepack)

# generalized linear model, but same as lm(time ~ route, data = baselong)
m.glm <- glm(time ~ route, family = gaussian(link = identity), data = baselong)

# generalized estimating equations
m.gee <- geeglm(time ~ route, family = gaussian(link = identity),
  id = player, corstr = "exchangeable", data = baselong)
```

Note: The data *must* be sorted by the `id` variable, and the `id` variable must be a *factor* or a *number* (not *character*). These data are already sorted, but if they were not we could use something like the following.

```
library(dplyr)
baselong <- baselong |> arrange(player)
```

Alternatively, without using the `dplyr` package, we could do this.

```
baselong <- baselong[order(baselong$player),]
```

Comparing inferences for the model parameters.

```
summary(m.glm)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.53409	0.0572	96.784	3.05e-70
routeround	0.00909	0.0809	0.112	9.11e-01
routerwide	-0.07500	0.0809	-0.927	3.57e-01

```
summary(m.gee)
```

Call:

```
geeglm(formula = time ~ route, family = gaussian(link = identity),
  data = baselong, id = player, corstr = "exchangeable")
```

Coefficients:

	Estimate	Std.err	Wald	Pr(> W)
(Intercept)	5.53409	0.05411	10461.38	< 2e-16 ***
routeround	0.00909	0.02564	0.13	0.72
routerwide	-0.07500	0.01839	16.63	4.6e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = exchangeable

Estimated Scale Parameters:

```
              Estimate Std.err
(Intercept)  0.0687  0.0278
Link = identity
```

Estimated Correlation Parameters:

```
              Estimate Std.err
alpha        0.896  0.0585
Number of clusters: 22 Maximum cluster size: 3
```

Comparing inferences for the expected time for each route.

```
library(emmeans)
```

```
emmeans(m.glm, ~route)
```

route	emmean	SE	df	lower.CL	upper.CL
narrow	5.53	0.0572	63	5.42	5.65
round	5.54	0.0572	63	5.43	5.66
wide	5.46	0.0572	63	5.34	5.57

Confidence level used: 0.95

```
emmeans(m.gee, ~route)
```

route	emmean	SE	df	lower.CL	upper.CL
narrow	5.53	0.0541	63	5.43	5.64
round	5.54	0.0566	63	5.43	5.66
wide	5.46	0.0568	63	5.35	5.57

Covariance estimate used: vbeta

Confidence level used: 0.95

Comparing inferences for the *differences* in expected time between routes.

```
pairs(emmeans(m.glm, ~route), adjust = "none", infer = TRUE)
```

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
narrow - round	-0.0091	0.0809	63	-0.1707	0.152	-0.112	0.9110
narrow - wide	0.0750	0.0809	63	-0.0866	0.237	0.927	0.3570
round - wide	0.0841	0.0809	63	-0.0775	0.246	1.040	0.3020

Confidence level used: 0.95

```
pairs(emmeans(m.gee, ~route), adjust = "none", infer = TRUE)
```

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
narrow - round	-0.0091	0.0256	63	-0.0603	0.0421	-0.350	0.7240
narrow - wide	0.0750	0.0184	63	0.0382	0.1118	4.080	<.0001
round - wide	0.0841	0.0307	63	0.0227	0.1455	2.740	0.0080

Confidence level used: 0.95

Note that the `contrast` function from **trtools** will also work here.

Example: Consider two approaches to the `smoking` data: ignoring the study effect entirely and a marginal model with inferences based on GEE.

```
head(quitsmoke)
```

```
# A tibble: 6 x 4
  study      treatment quit total
  <fct>      <chr>    <int> <int>
1 Blondal89   gum         37    92
2 Blondal89   control     24    90
3 Campbell191 gum         21   107
4 Campbell191 control     21   105
5 Fagerstrom82 gum         30    50
6 Fagerstrom82 control     23    50

m.glm <- glm(cbind(quit, total - quit) ~ treatment,
  family = binomial, data = quitsmoke)
m.gee <- geeglm(cbind(quit, total - quit) ~ treatment,
  family = binomial, data = quitsmoke,
  id = study, corstr = "exchangeable")
```

Comparing inferences for the model parameters.

```
summary(m.glm)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.450	0.0490	-29.59	1.76e-192
treatmentgum	0.507	0.0631	8.04	9.11e-16

```
summary(m.gee)
```

Call:

```
geeglm(formula = cbind(quit, total - quit) ~ treatment, family = binomial,
  data = quitsmoke, id = study, corstr = "exchangeable")
```

Coefficients:

	Estimate	Std.err	Wald	Pr(> W)
(Intercept)	-1.444	0.116	155.5	< 2e-16 ***
treatmentgum	0.501	0.078	41.2	1.4e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = exchangeable

Estimated Scale Parameters:

	Estimate	Std.err
(Intercept)	0.0601	0.0158

Link = identity

Estimated Correlation Parameters:

	Estimate	Std.err
alpha	0.445	0.229

Number of clusters: 26 Maximum cluster size: 2

Estimating the probability of quitting.

```
emmeans(m.glm, ~treatment, type = "response")
```

treatment	prob	SE	df	asyp.LCL	asyp.UCL
-----------	------	----	----	----------	----------

control	0.19	0.00754	Inf	0.176	0.205
gum	0.28	0.00801	Inf	0.265	0.296

Confidence level used: 0.95

Intervals are back-transformed from the logit scale

```
emmeans(m.gee, ~treatment, type = "response")
```

treatment	prob	SE	df	lower.CL	upper.CL
control	0.191	0.0179	50	0.158	0.230
gum	0.280	0.0255	50	0.232	0.334

Covariance estimate used: vbeta

Confidence level used: 0.95

Intervals are back-transformed from the logit scale

Estimating the odds ratio for the effect of the gum treatment.

```
pairs(emmeans(m.glm, ~treatment, type = "response"),
      reverse = TRUE, infer = TRUE)
```

contrast	odds.ratio	SE	df	asympt.LCL	asympt.UCL	null	z.ratio	p.value
gum / control	1.66	0.105	Inf	1.47	1.88	1	8.040	<.0001

Confidence level used: 0.95

Intervals are back-transformed from the log odds ratio scale

Tests are performed on the log odds ratio scale

```
pairs(emmeans(m.gee, ~treatment, type = "response"),
      reverse = TRUE, infer = TRUE)
```

contrast	odds.ratio	SE	df	lower.CL	upper.CL	null	t.ratio	p.value
gum / control	1.65	0.129	50	1.41	1.93	1	6.420	<.0001

Confidence level used: 0.95

Intervals are back-transformed from the log odds ratio scale

Tests are performed on the log odds ratio scale

Example: Consider two approaches to the Sitka data.

```
m.glm <- glm(treesize ~ Time * treat,
  family = gaussian(link = identity), data = Sitka)
m.gee <- geeglm(treesize ~ Time * treat,
  family = gaussian(link = identity), data = Sitka,
  id = tree, corstr = "exchangeable")
```

Comparing inferences for the model parameters.

```
summary(m.glm)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-305.123	52.711	-5.79	1.46e-08
Time	2.509	0.256	9.80	2.03e-20
treatozone	110.675	63.755	1.74	8.34e-02
Time:treatozone	-0.788	0.310	-2.54	1.13e-02

```
summary(m.gee)
```

```
Call:
geeglm(formula = treesize ~ Time * treat, family = gaussian(link = identity),
       data = Sitka, id = tree, corstr = "exchangeable")
```

Coefficients:

	Estimate	Std.err	Wald	Pr(> W)
(Intercept)	-305.123	32.737	86.87	<2e-16 ***
Time	2.509	0.264	90.62	<2e-16 ***
treatozone	110.675	38.775	8.15	0.0043 **
Time:treatozone	-0.788	0.306	6.62	0.0101 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = exchangeable

Estimated Scale Parameters:

	Estimate	Std.err
(Intercept)	11432	2036

Link = identity

Estimated Correlation Parameters:

	Estimate	Std.err
alpha	0.752	0.0189

Number of clusters: 79 Maximum cluster size: 5

Estimating the growth rate in each treatment condition.

```
pairs(emmeans(m.glm, ~Time|treat, at = list(Time = c(250,150))))
```

treat = control:

contrast	estimate	SE	df	t.ratio	p.value
Time250 - Time150	251	25.6	391	9.800	<.0001

treat = ozone:

contrast	estimate	SE	df	t.ratio	p.value
Time250 - Time150	172	17.4	391	9.880	<.0001

```
pairs(emmeans(m.gee, ~Time|treat, at = list(Time = c(250,150))))
```

treat = control:

contrast	estimate	SE	df	t.ratio	p.value
Time250 - Time150	251	26.4	391	9.520	<.0001

treat = ozone:

contrast	estimate	SE	df	t.ratio	p.value
Time250 - Time150	172	15.6	391	11.030	<.0001

Comparing the growth rates between the treatment conditions.

```
pairs(pairs(emmeans(m.glm, ~Time|treat, at = list(Time = c(250,150)))), by = NULL)
```

contrast	estimate	SE	df	t.ratio	p.value
(Time250 - Time150 control) - (Time250 - Time150 ozone)	78.8	31	391	2.544	0.0113

```
pairs(pairs(emmeans(m.gee, ~Time|treat, at = list(Time = c(250,150)))), by = NULL)
```

contrast	estimate	SE	df	t.ratio	p.value
(Time250 - Time150 control) - (Time250 - Time150 ozone)	78.8	30.6	391	2.573	0.0105

Estimating expected sizes at 150, 200, and 250 days, as well as the differences between the ozone conditions.

```
emmeans(m.glm, ~treat|Time, at = list(Time = c(150,200,250)))
```

Time = 150:

treat	emmean	SE	df	lower.CL	upper.CL
control	71	16.50	391	38.8	104
ozone	64	11.20	391	41.7	86

Time = 200:

treat	emmean	SE	df	lower.CL	upper.CL
control	197	9.63	391	177.8	216
ozone	150	6.55	391	136.9	163

Time = 250:

treat	emmean	SE	df	lower.CL	upper.CL
control	322	15.50	391	291.7	353
ozone	236	10.60	391	215.1	257

Confidence level used: 0.95

```
emmeans(m.gee, ~treat|Time, at = list(Time = c(150,200,250)))
```

Time = 150:

treat	emmean	SE	df	lower.CL	upper.CL
control	71	10.8	391	50.1	92
ozone	64	4.7	391	54.5	73

Time = 200:

treat	emmean	SE	df	lower.CL	upper.CL
control	197	22.2	391	153.1	240
ozone	150	11.3	391	127.5	172

Time = 250:

treat	emmean	SE	df	lower.CL	upper.CL
control	322	34.9	391	253.6	391
ozone	236	18.9	391	198.7	273

Covariance estimate used: vbeta

Confidence level used: 0.95

```
pairs(emmeans(m.glm, ~treat|Time, at = list(Time = c(150,200,250))), infer = TRUE)
```

Time = 150:

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
control - ozone	7.5	20.0	391	-31.7	46.8	0.380	0.7060

Time = 200:

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
control - ozone	46.9	11.6	391	24.0	69.8	4.030	<.0001

Time = 250:

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
control - ozone	86.3	18.8	391	49.4	123.3	4.600	<.0001

Confidence level used: 0.95

```
pairs(emmeans(m.gee, ~treat|Time, at = list(Time = c(150,200,250))), infer = TRUE)
```

Time = 150:

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
control - ozone	7.5	11.7	391	-15.54	30.6	0.642	0.5210

Time = 200:

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
control - ozone	46.9	24.9	391	-2.05	95.9	1.884	0.0600

Time = 250:

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
control - ozone	86.3	39.7	391	8.36	164.3	2.177	0.0300

Confidence level used: 0.95

Limitations of Marginal Models and GEE

1. Performs best when the data are relatively “shallow” meaning that there are many units (e.g., players, studies, or trees) but relatively few observations per unit (e.g., routes, treatment conditions, time points).
2. Inefficient if the (working) correlation structure is a poor approximation.
3. Limited to “marginal inferences” in that it cannot tell us anything about the variation among units (in contrast to models with “random effects” which we will discuss later).